



Society of Petroleum Engineers

SPE-230237-MS

MIC Risk Assessment in Upstream Pipelines: An Optimized Corrosion Mitigation Approach

M. Al Seiri, M. Al Hammadi, and N. Al Khyeli, ADNOC Onshore; M. Al Jaber, ADNOC Offshore; M. Mohamed-Said, R. Quintane, A. Silva, and H. Marchebois, TotalEnergies

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/230237-MS](https://doi.org/10.2118/230237-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

Microbiologically influenced corrosion (MIC), primarily driven by the presence of sulfate-reducing bacteria (SRB), poses a continuous threat to the integrity of pipeline networks. An MIC-prone environment is characterized by high levels of nutrients (such as sulfates, nitrates, and organic acids), low temperatures, low flow velocities or stagnant flow regimes, and a high bacterial cell count. In upstream operations, existing mitigation efforts include pigging, biocide batch treatment, and continuous serial dilution test methods for bacterial monitoring. To assess the existing microbial risk, a comprehensive assessment was conducted and benchmarked against peer upstream operators and the industry's best practices. A multifactorial risk model was applied, incorporating nutrient availability, flow velocity and regime, temperature, microbial activity, and hydrogen sulfide (H₂S) partial pressure as thresholds for SRB auto-toxicity. By utilizing this data-driven framework, the pipelines were successfully classified into low- and high-risk MIC categories.

The study also presents evaluations of the MIC risk model, as well as a supported case studies based on historical field data and In-line Inspection (ILI) survey results. Using the results of these classifications, an improved management strategy was introduced by reducing biocide dosage frequency in low-risk segments and increasing kill contact time in high-risk segments. Various types of active monitoring were recommended as part of the optimization program, including inlet/outlet bacterial counts, bio-probes, and corrosion coupons to monitor microbially induced corrosion, as well as inspection data and DNA-based microbial analysis. This targeted approach resulted in a reduction of biocide usage by more than 80%, leading to significant annual cost savings, a reduction in OPEX, and improvements in pipeline integrity management. This integrated methodology demonstrates the effectiveness of data-driven risk assessment and tailored treatment strategies, establishing a sustainable and economically viable benchmark for MIC management in upstream pipeline operations.

Introduction

Microbiologically influenced corrosion (MIC) represents a localized and aggressive form of internal corrosion that poses a distinct and severe threat to the structural integrity of pipelines and storage tanks. In contrast to uniform corrosion, which progresses predictably and uniformly across metal surfaces, MIC

is spatially heterogeneous and is frequently undetectable by conventional monitoring methodologies. MIC primarily occurs in oil/water systems but is not a common cause of failure in gas pipelines; corrosion rates typically range from 0.5 to 2 mm/year (≈ 78.74 MPY). At worst, rates can reach 20 mm/year (≈ 787.40 MPY), which could lead to pipeline failure within 1 to 2 years if not effectively managed. MIC is typically detected in conjunctions with under-deposit corrosion, particularly in low-flow, stagnant regions or pipeline dead legs where pigging operations are restricted (Bruijnen, 2020).

MIC phenomenon primarily arises from the metabolic activities of sulfidogenic microorganisms, most notably sulfate-reducing bacteria, which alter the local electrochemical factors and corrosion protection layers. With MIC, corrosion is often caused by bacteria attaching to and forming a biofilm on metal surfaces, leading to micro-zones of conditions highly conducive to pitting. Sulfide films often act as initiators of MIC, though their presence does not invariably correlate with active microbial colonization (King, 1972); (Lee, 1993).

The complexity of MIC is underscored by its overlap with abiotic corrosion mechanisms. Weak acids, such as hydrogen sulfide (H_2S), carbon dioxide (CO_2), acetic acid ($\text{CH}_3\text{CO}_2\text{H}$), and water (H_2O), are central to both abiotic corrosion and microbial metabolism. Their corresponding anions (HS^- , HCO_3^- , Ac^- , OH^-) play multiple roles: they act as buffering agents, serve as surface adsorbates in electrochemical reactions, and form soluble or precipitated corrosion products (e.g., FeS , FeCO_3 , $\text{Fe}(\text{OH})_2$). This intricate interplay of local chemistry, transport, electrochemistry, and precipitation is further complicated in environments where multiple acids coexist, such as within microbial biofilms (Crolet, 2005). Therefore, understanding abiotic corrosion is essential for interpreting MIC. The metabolic byproducts of sulfidogenic bacteria (e.g., H_2S , CO_2 , acetate) mimic those already present in oilfield systems, making it challenging to distinguish microbial from purely chemical corrosion (Crolet, 2005); (Bonis, 2015). This reinforces the idea that MIC is not purely "microbially induced," but rather microbially influenced, superimposed on an abiotic baseline.

As for sulfidogenic microorganisms, the term "sulfidogenic bacteria" is primarily a chemical classification based on the nature of their metabolic outputs, namely sulfide, carbonate, and acetate, rather than a strict taxonomic or microbiological definition. These metabolic functions are found across a broad spectrum of microbial species and genera, increasing the likelihood of their presence in oilfield environments. Notably, these bacteria include both sulfate-reducing bacteria (SRB) and thiosulfate-reducing bacteria (TRB), acid-producing bacteria, as well as certain Archaea. Sulfate reduction occurs when organic matter or hydrogen serves as the electron donor, and sulfate acts as the terminal electron acceptor. Depending on their metabolic pathways, SRBs may either completely oxidize organic matter to carbon dioxide or incompletely oxidize it, leaving acetic acid as a byproduct. The resulting metabolites H_2S , H_2CO_3 , and Hac contribute directly to corrosion processes by altering the local pH and redox conditions beneath biofilms. (Crolet, 2005). Despite being obligatory anaerobes, SRB can survive in oxygenated environments, existing both as sessile biofilm communities and planktonic cells. They tolerate a broad pH range (5.0–9.5) and variable salinity levels, with MIC risk persisting in brines exceeding 300 g/L (NACE 31205, 2006). SRB can grow optimally between 25 and 45 °C, with some thermophilic strains metabolizing at temperatures greater than 60 °C (NACE 31205, 2006). In contrast, H_2S at concentrations exceeding 547 mg/L has a reversible inhibitory effect on SRB metabolism. These metabolic traits are common to other sulfate-reducing microorganisms and include other thiosulfate-reducing bacteria and several Archaea, all of which contribute to corrosion through the production of H_2S , carbonic acid (H_2CO_3), and acetic acid, thus modulating local pH and redox conditions (Crolet, 2005); (Kermani Bijan, 2019).

The morphologies of MIC, including grainy surfaces with very fine micro-pitting, dense biofilm, and subsurface tunnelling produced by the dissolution of manganese sulfide (Crolet, 2005), highlight the need for comprehensive analysis. However, morphology alone cannot provide a definitive distinction between MIC and other types of corrosion. This underlines the requirement for in-depth analysis conducted in an integrated manner, which includes morphological assessment as well as monitoring of microbial

mechanisms. The reporter detects changes in local chemistry beneath biofilms, not just surface morphology. The characteristics of MIC are depicted in Figure 1, and a schematic diagram, shown in Figure 2, describes the essential conditions for MIC formation, its mechanism, and biofilm formation.



Figure 1—Typical features of MIC type damage characteristics of localized morphologies (Kermani Bijan, 2019).

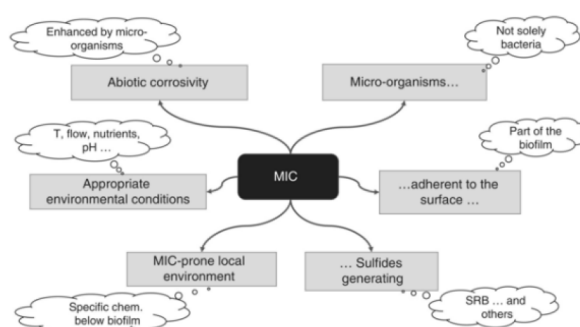
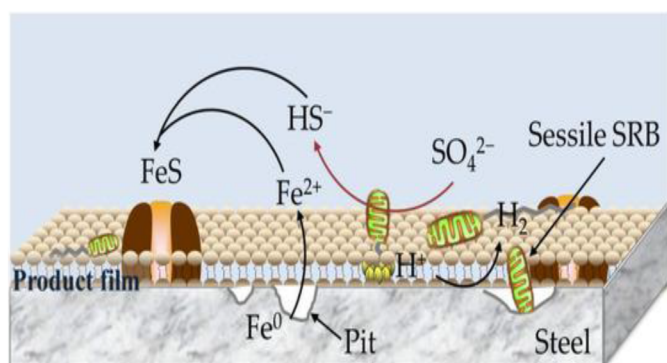


Figure 2—Schematic of the key conditions required for MIC occurrence (Kermani Bijan, 2019) and mechanism of biofilm formation (Chen, 2021).

Mitigation strategies encompass biocide application alongside mechanical cleaning (pigging). Best practices recommend alternating biocides with distinct modes of action, such as tetrakis(hydroxymethyl) phosphonium sulfate (THPS) and protein-denaturing agents (e.g., glutaraldehyde), to enhance efficacy (SP0106, 2018). The effectiveness of biocides is influenced by parameters including dosage, contact time, phase partitioning, chemical stability, and compatibility, with active ingredient concentrations commonly reaching up to 50%. Robust microbial control necessitates systematic pre- and post-treatment sampling, considering system variables such as water content, flow dynamics, solids concentration, temperature, and chemical composition. Recent advancements in microbial monitoring, such as routine adenosine triphosphate (ATP) assays, coupled with molecular techniques like quantitative polymerase chain reaction (qPCR) and metagenomic analyses (TM0194, 2014), underscore the importance of these techniques. These methodologies facilitate detailed microbial community profiling, thereby informing adaptive biocide regimes that optimize microbial control while balancing operational expenditures and maintaining pipeline integrity.

Operational Context and Challenges in Managing Microbiologically Influenced Corrosion (MIC)

In upstream production networks, MIC mitigation is typically executed through standardized protocols applied across the asset base (Figure 3). For piggable pipelines, standard practice involves biocide dosing at 500 ppm (based on total liquid) for three hours, aligned with pigging schedules, and alternating between two

biocide formulations. Core strategies include mechanical pigging to remove deposits and disrupt biofilms, batch biocide injection to target microbial populations, and periodic chemical cleaning of pipelines and associated infrastructure.

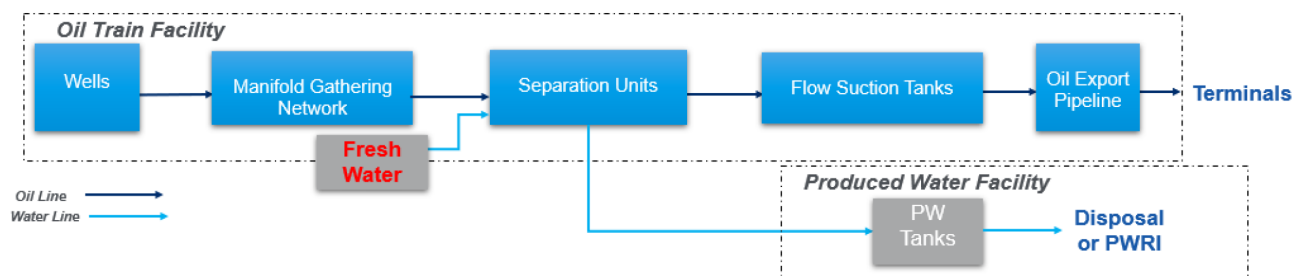


Figure 3—Entire Production Network and Exiting Biocide Treatment

Traditional corrosion monitoring techniques, while effective for detecting uniform corrosion, lack sensitivity for localized MIC. Even advanced methods are more adept at detecting pit growth than at identifying early microbial colonization or pit initiation. This underscores the urgent need for improved MIC detection and management strategies, which require integration with microbiological monitoring to enable early detection, optimize treatment intervals, and enhance the efficacy of biocide application (Crolet, 2005).

Despite their general effectiveness, current mitigation practices are often applied uniformly or reactively, which can be seen as a constraint. This approach introduces key limitations and underscores the need for change and innovation in the field of corrosion control:

1. **Under-treatment** accelerates corrosion and increases the risk of premature failure.
2. **Over-treatment** elevates operational costs and environmental impact due to excessive chemical use.
3. **Uniform dosing** neglects localized microbial activity and site-specific risk profiles.
4. **Low adaptability** to dynamic operational conditions such as salinity, flow velocity, and temperature can result in suboptimal treatment and an increased risk of corrosion. For example, changes in salinity levels can influence the growth of certain bacteria, while variations in flow velocity can affect the distribution of biocides.
5. **Insufficiently different microbiological detection** (e.g., SRB, TRB, NRB counts) limits the use of adaptive, data-driven dosing strategies.

Given the resilience of sulfate-reducing bacteria (SRB) and thiosulfate-reducing bacteria (TRB), the application of robust microbial monitoring techniques, combined with accurate sampling locations, is crucial for precise diagnostics. Overcoming these challenges requires a shift toward targeted, responsive, and diagnostics-driven MIC management. This approach is not only beneficial but also necessary for enhancing operational efficiency, minimizing environmental impact, and maintaining asset integrity.

Pipeline Pigging and Biocide Application Strategy

A pipeline inspection gauge, commonly referred to as a pig, is an essential device designed for the effective cleaning and thorough inspection of pipelines. The term originates from early pigs bundles of straw wrapped in wire or leather that produced a squealing sound when pushed through pipes. Modern pigs are typically spherical or cylindrical in shape and vary in terms of material, size, color, and accessory features. All pigs closely match the internal diameter of the pipeline allows to remove debris and detect cracks or flaws effectively.

Pigging is critical maintenance practice throughout the pipeline lifecycle from pre-commissioning cleaning and routine maintenance to corrosion control, rehabilitation, and decommissioning. Common pig types also include bi-directional pigs equipped with brushes or scrapers for removing wax and debris, as well

as general-purpose cylindrical or spherical pigs. Smart pigs are deployed periodically for detailed inspection and long-term monitoring (El-Sherik, 2017). Figure 4 illustrates a pig in upstream operations, followed by a list of pigs and their functions.

- Utility pigs (foam, brush, cup, scraper): Remove wax, debris, and scale to maintain flow efficiency.
- Gauging pigs: Detect internal restrictions, dents, or deformations.
- Batching pigs: Separate fluids in multiproduct pipelines to prevent cross-contamination.
- Smart pigs: Advanced inspection tools for integrity assessment, including:
 - Magnetic Flux Leakage (MFL): Detect internal and external metal loss.
 - Ultrasonic Testing (UT): Measure wall thickness and identify cracks or laminations.
 - Caliper pigs: Record internal diameter changes, dents, and ovalities.
 - Electromagnetic Acoustic Transducer (EMAT): Detect cracks and stress corrosion without a liquid couplant.
 - Hybrid pigs: Combine multiple technologies (e.g., MFL + caliper or UT + EMAT) for comprehensive inspection.
- Specialty pigs (gel, sphere, dual-diameter): Perform niche functions such as dewatering, chemical application, or handling variable pipeline diameters.

Biocide Application Strategy and Integrated Benefits

Integrating mechanical pigging with biocide treatment provides significant operational and integrity benefits. This approach improves microbial control, mitigates microbiologically influenced corrosion (MIC), enhances flow assurance by removing biofilms and deposits, extends pipeline service life, optimizes operational costs, and reinforces overall pipeline reliability. Such a combined mechanical-chemical strategy is widely recognized as best practice in the industry for long-term asset protection while minimizing unplanned interruptions (El-Sherik, 2017).

Pigging physically removes deposits and biofilms, exposing fresh metal surfaces and enabling more effective biocide penetration. Best-in-class programs typically apply biocides post-pigging to maximize chemical contact. Biocide type, dosage, and treatment frequency are determined based on water chemistry, microbial monitoring data, and operational requirements (Tiratsoo, 2013). When implemented according to established protocols, this integrated approach effectively mitigates corrosion risks, maintains flow assurance, prolongs pipeline life, and strengthens overall system integrity. The combination of mechanical cleaning and chemical treatment remains a foundation of pipeline maintenance and integrity management strategies.

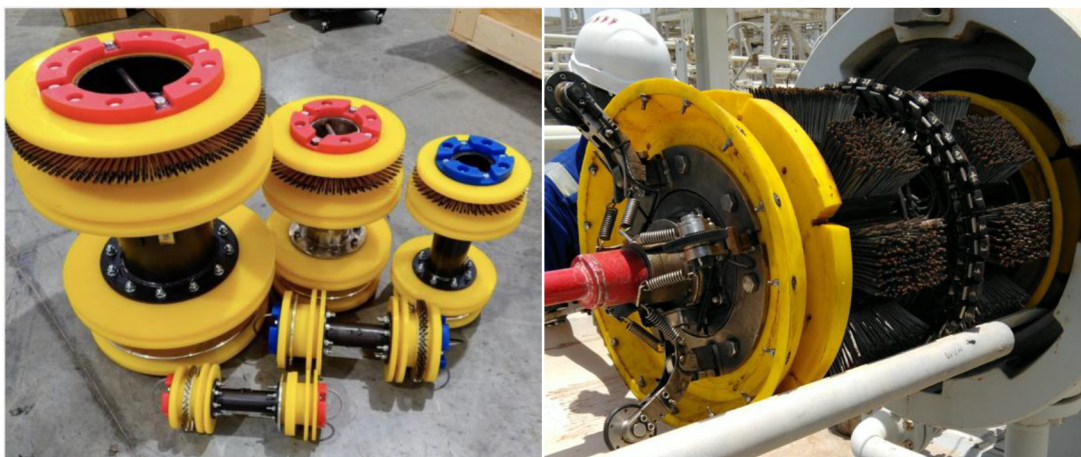


Figure 4—Illustration of Pig Devices Utilised in Upstream Operations (Bi-Directional Disc Pig shown on the left, Magnetic Flux Leakage [MFL] Pig depicted on the right)

Study Aim and Strategic Approach

This study presents a risk-based management framework for MIC in upstream pipeline networks, designed to address the shortcomings of conventional and mitigation approaches. The framework integrates historical integrity records, which include past instances of microbial-induced corrosion and the measures taken, as well as operational data, such as flow rates, water cut, H_2S concentration, benchmarked against key MIC indicators from peer operators and internationally recognized best practices.

This supports customized and targeted biocide application treatments based on quantified microbial risk, along with advanced microbial diagnostics for early biofilm detection, and dynamic dosing strategies responsive to real-time microbial activity and evolving field conditions, such as changes in temperature, pressure, or chemical composition. The enhanced surveillance program incorporates monitoring of Sulfate-, Thiosulfate-, and Nitrate-Reducing Bacteria (SRB, TRB, NRB), DNA-based microbial identification, and optimized sampling protocols to ensure diagnostic precision.

This data-based methodology marks a strategic transition toward predictive, data-driven integrity management. The framework's objectives include extending asset service life, reducing operational expenditure, lowering chemical consumption, and mitigating environmental impact. The significant reduction in operational spending is a key benefit that the audience can look forward to. By embedding integrated microbial and corrosion diagnostics into decision-making, the framework establishes a technical benchmark for MIC control in upstream pipeline systems, particularly in environments influenced by sulfidogenic microorganisms.

Peer-Reviewed Framework in MIC Risk Assessment Development

To enhance predictive integrity management within upstream operations, a specialized MIC risk assessment framework has been developed for pipeline systems. This framework has been thoroughly benchmarked against global industry standards and operational insights from upstream peer operators, to ensure alignment with best practices and practical field conditions. The approach is data-centric, incorporating microbial diagnostic tools, fluid chemistry profiles, and detailed flow regime analyses to specifically evaluate internal corrosion in carbon steel assets, including pipelines, pressure vessels, and storage tanks. While MIC can also affect stainless steel under certain aerobic or chemically unique environments with different microbial populations, these scenarios are beyond the scope of the current framework.

The model focuses on key MIC agents, primarily sulfate-reducing bacteria (SRB) and thiosulfate-reducing bacteria (TRB). Although recent studies suggest possible involvement of methanogenic archaea under elevated temperature conditions and acid-producing bacteria within mixed microbial communities,

these organisms have not yet to be conclusively demonstrated as primary contributors to MIC in the asset base under consideration and are therefore omitted pending further validation.

This comprehensive and integrated model ensures a balance between scientific accuracy and operational applicability. The semi-quantitative risk classification scheme, adapted from Peer Upstream Operator methodology, organizes critical environmental and operational variables into graded risk categories, reflecting the combined influence of these factors on MIC susceptibility.

Critical factors influencing MIC risk encompass:

- **Temperature:** The risk is lowest outside the microbial growth range ($<15\text{ }^{\circ}\text{C}$ or $>100\text{ }^{\circ}\text{C}$), increases moderately above $60\text{ }^{\circ}\text{C}$, and peaks within the $15\text{--}60\text{ }^{\circ}\text{C}$ range, which is conducive to microbial activity.
- **Sulfate and Thiosulfate Levels:** Elevated sulfate ($>50\text{ mg/L}$) and thiosulfate ($>1\text{ mg/L}$) concentrations substantially raise the potential for MIC, with even trace thiosulfate amounts critically supporting thiosulfate-reducing bacteria.
- **Water sources, such as reservoirs** and seawater, due to their rich microbial ecosystems and nutrient loads, represent the highest risk. Utility water corresponds to an intermediate risk level, while conditions of condensed, fresh, or absent water correlate with lower MIC risk.
- **Flow Conditions:** Low or stagnant flow ($<1\text{ m/s}$), especially in dead zones, encourages biofilm development and microbial colonization. Conversely, high-velocity flows ($>3\text{ m/s}$), such as annular or slug flows, generally mitigate risk, while moderate flows ($1\text{--}3\text{ m/s}$), particularly with stratification, pose intermediate risk.
- **Hydrogen Sulfide Concentration:** H_2S partial pressure above 200 mbar may inhibit SRB activity, thereby reducing the MIC risk; however, this effect is context-dependent and not definitive in all scenarios.
- **Microbial Quantification:** Presence of SRB and TRB populations exceeding 100 cells/cm^3 is a clear indicator of elevated MIC risk.

Final MIC risk ratings are derived by synthesizing individual parameter assessments, with particular consideration given to the co-occurrence of multiple high-risk factors. This integrative evaluation offers adaptability to diverse field data conditions and operational environments, facilitating precise MIC risk profiling and enabling more focused and effective corrosion mitigation interventions in upstream oil and gas infrastructure.

Pipeline MIC Risk Assessment: Benchmarking Against Peer Operators and Case Study Evaluation

A comprehensive benchmarking study was conducted on Microbiologically Influenced Corrosion (MIC) Risk Model and key parameters, which were calibrated against extensive historical operational and integrity data from upstream pipeline assets spanning the period from 2020 to 2024 as shown in [Table 1](#) and [Figure 5](#). This comparative evaluation facilitated the systematic stratification of MIC risk levels across the pipeline network, enabling the prioritization of inspection, monitoring, and mitigation activities. Such stratification enhances operational efficiency, reduces unnecessary expenditures, and supports targeted integrity management within MIC-prone environments.

Table 1—MIC Risk Assessment Results Benchmarked with Peer's Operator

Category *	Transfer Lines	MOL	Peer's Operator MIC Risk Model
Temperature	100°C to 60°C	< 60°C	<15°C or >100°C
Nutrients (i.e Sulfate content)	20–300 mg/l	20–300 mg/l	> 5 mg/l
Salinity	250,000 mg/l	12,000 mg/l	NA
Water Cut%	>70%	<1%	NA
H ₂ S partial pressure	>0.2 bar	No H ₂ S	> 0.2 bar
H ₂ S & CO ₂ mol%	Field A: H ₂ S= 0 mol% and CO ₂ =5 mol% Field B: H ₂ S=15 mol% and CO ₂ = 8 mol%	H ₂ S= 0.05 mol% CO ₂ = 0.05 mol%	NA
Flow Velocity	> 3 m/s	1–2 m/s	> 3 m/s
Bacterial Monitoring	No more than 100 Cells/ml	More than 100 Cells/ml	No more than 100 Cells/ml
Score	Low Risk	High Risk	Low Risk

* "No Risk- Low Risk" = All values within MIC Risk Model acceptable limits. "High Risk" = One or more values exceed limits.

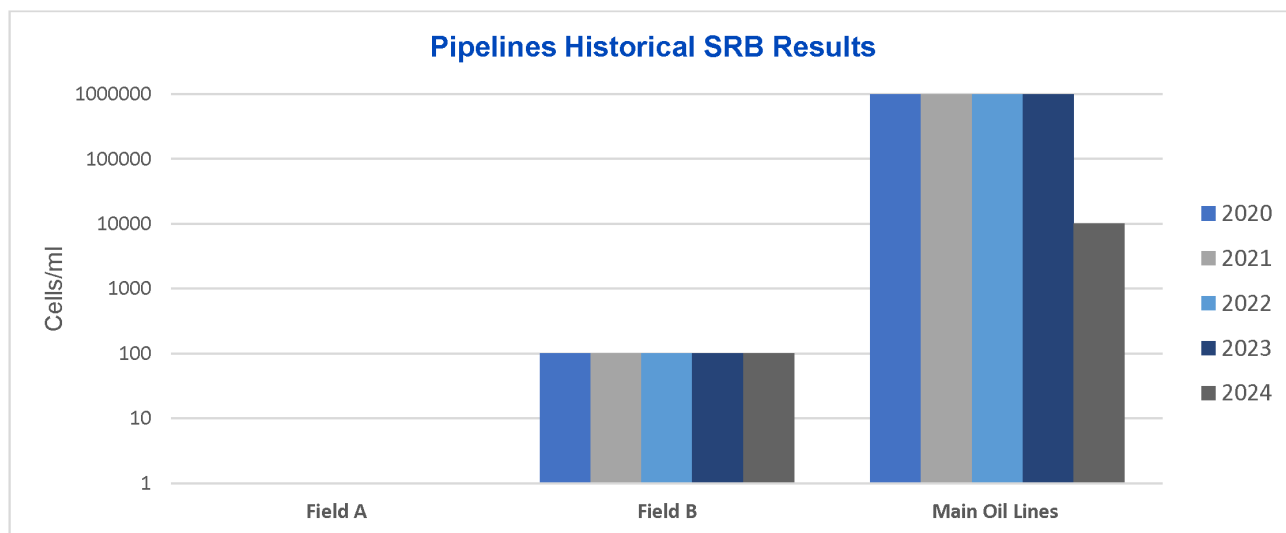


Figure 5—Pipeline Historical SRB Serial Dilution Results

The risk assessment primarily focuses on sulfate-reducing bacteria (SRB) due to their well-documented role as principal MIC agents. Key parameters integrated into the evaluation included detailed water chemistry metrics, such as sulfate, chloride, and bicarbonate concentrations, as well as redox potential and pH, alongside microbial activity data associated explicitly with SRB populations. The resulting MIC risk classifications are presented in the appended risk matrix, which serves as a robust, data-driven tool for proactive management of corrosion.

Supporting the matrix evaluation, technical mapping of SRB populations was conducted across multiple fields over five years of historical data. Notably, Field A exhibited an absence of detectable SRB, attributable to the lack of hydrogen sulfide (H₂S) in the fluid medium. Conversely, Field B, characterized by elevated H₂S concentrations, initially reported false-positive SRB presence attributed to contamination during sampling; these results were subsequently invalidated by quantitative polymerase chain reaction (qPCR) analyses, which confirmed the absence of SRB. Other fields demonstrated detectable SRB populations that decreased over time, a trend attributed to both H₂S interference in microbial assays and strategic increases in biocide concentration from 20% to 50%. This underscores the critical importance of maintaining and verifying biocide concentrations to ensure sustained microbial control efficacy.

Operationally, transfer lines within these fields operate under high-temperature and high-flow velocity conditions, which inherently suppress MIC risk. Consequently, biocide treatment regimens in these segments were optimized to reflect the relatively low susceptibility of the target organisms.

The application of the MIC Risk Matrix across pipeline infrastructure revealed a clear stratification of corrosion susceptibility, allowing for refined integrity management approaches, as follows:

- **Low-Risk Pipelines (e.g., Transfer Lines):** Characterized by continuous fluid flow, elevated flow velocities, high H₂S concentration, and limited microbial colonization. These factors collectively inhibit biofilm development and microbial proliferation, thereby markedly reducing MIC risk.
- **High-Risk Pipelines (e.g., Main Oil Lines - MOLs):** Characterized by stagnant or intermittent flow regimes, lower salinity in produced water, nutrient-rich conditions, and persistent SRB activity. Complex pipeline geometries and flow disturbances can contribute to increased microbial colonization and localized corrosion susceptibility. Two MOLs were identified as low risk due to the absence of reported MICs, a finding supported by IPS Inspection data, which will be discussed in the following case study section.

Technical Considerations in SRB Behavior and Sampling

- **SRB Self-Inhibition:** Elevated concentrations of hydrogen sulfide (H₂S) may suppress SRB metabolic activity, potentially masking the presence of SRB during routine assessments.
- **Sampling Post-Pigging:** Samples collected immediately after pigging may contain dislodged biofilms or debris from non-representative locations, leading to false-positive SRB detections or inflated counts.
- **Validation Through Biomapping and DNA Level qPCR:** Discrepancies observed between post-pigging samples and routine microbial monitoring were verified supported by qPCR. This is confirming the importance of systematically appropriate sampling and preservation protocols.

This stratified MIC risk classification framework enabled to prioritize inspection and mitigation resources on the most vulnerable pipeline segments. The transition toward this data-driven, risk-based approach has resulted in corrosion control programs that are more precise, operationally efficient, and aligned with international best practices. Furthermore, the MIC Risk Matrix exhibits strong scalability and adaptability, enabling its application across various upstream production environments with differing operational conditions and microbial challenges.

Case Study: MIC Risk Matrix Application in a Low-Risk Pipeline

A stabilized main oil line (MOL), commissioned in 2012 and inspected in 2021 and 2023, was evaluated using recent inline inspection (ILI) data and classified as *Low Risk* under the MIC Risk Model. The pipeline data is summarized in [Table 2](#). During its initial years of operation, the pipeline functioned without biocide treatment. Following the introduction of biocide dosing, subsequent internal pipe surface (IPS) evaluations and comparative ILI results confirmed the absence of microbiologically influenced corrosion (MIC).

Table 2—Pipeline Data

Parameter	Value
Pipeline	Main Oil Line, Crude, Stabilized, Onshore
Diameter	48 inch
Length	147 km
Type of Pipe	Longitudinally weld pipe
Steel Grade	API 5L X65
Nominal Wall thickness	14/18/25 mm
Anomaly assessment code	ASME B31G
Coating Type	Metallic, no internal lining/Three-Layer PE or PP /Heat shrink sleeves
Buried Status	Buried
Corrosion Allowance	2 mm
Design Pressure	90 Bar
MAOP	73 Bar
Year of Commission	2012
Inspection History	2021, 2023

Magnetic Flux Leakage (MFL) ILI revealed metal loss predominantly attributable to external corrosion and mechanical defects, rather than MIC (Figs. 6–7; Tables 2–3). These findings, corroborated by sulfate-reducing bacteria (SRB) counts, provided further assurance of the pipeline's microbiological integrity. Notably, similar MOL pipelines installed in the 1980s, which operated with limited or no chemical treatment, have remained in service with controlled corrosion, underscoring the importance of continuous and integrated monitoring.

Table 3—Analysis Between IPS Results

Depth Range	Internal Anomalies (2023)	External Anomalies (2023)	Internal Anomalies (2021)	External Anomalies (2021)
30% ≤ metal loss < 80%	-	-	-	-
20% ≤ metal loss < 30%	1	-	1	-
10% ≤ metal loss < 20%	58	188	31	112
Metal loss < 10%	1571	3797	1529	3825
Total Number of Anomalies	1630	3985	1561	3937
Deepest Anomaly	29%	18%	29%	18%

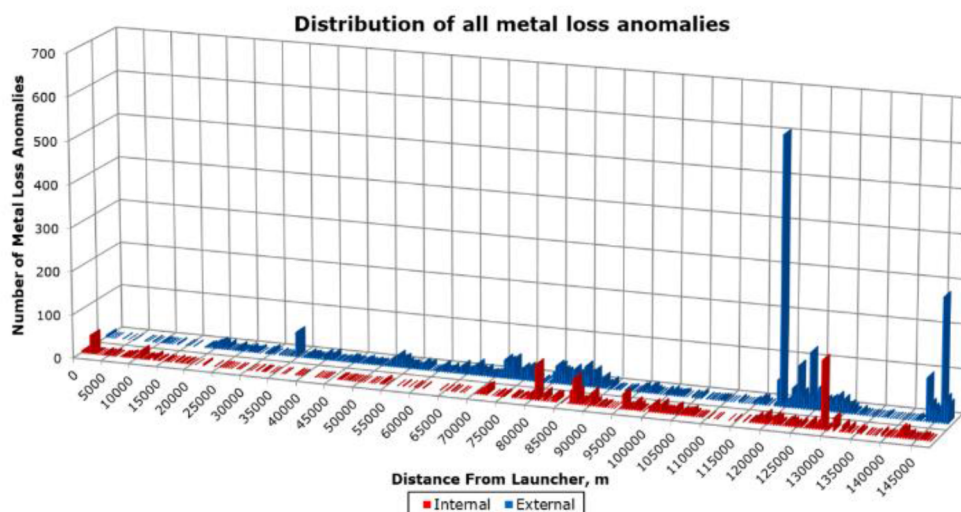


Figure 6—IPS results showing metal loss anomalies for Year 2021

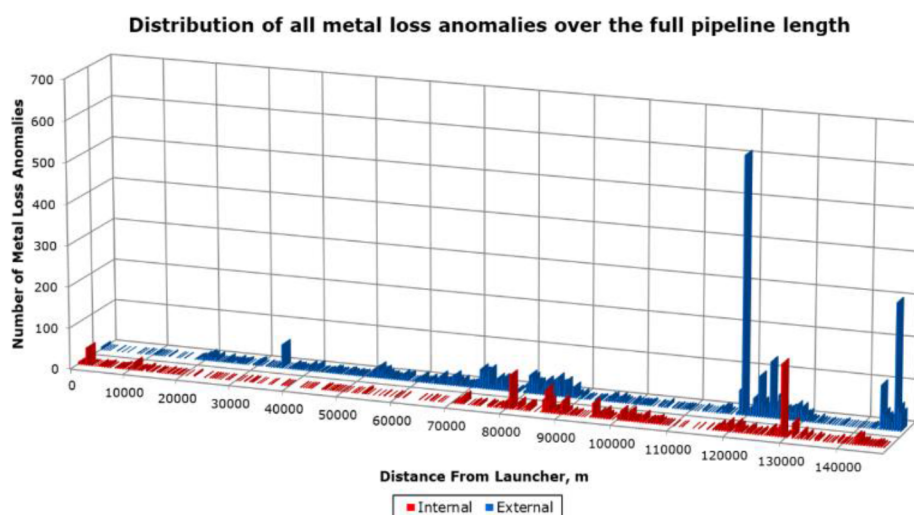


Figure 7—IPS results showing metal loss anomalies for Year 2023

In response, the implemented targeted mitigation measures include optimized biocide dosing strategies, revised pigging schedules, and an enhanced microbial monitoring protocol. This case study demonstrates the utility of the MIC Risk Model in translating microbiological risks into risk-informed corrosion management strategies. By integrating operational data, microbial diagnostics, and historical corrosion records, the model supports predictive MIC mitigation as part of a comprehensive integrity management framework.

Biocide Treatment Optimization Strategy for MIC Control in Upstream Pipelines

A data-driven and risk-based biocide optimization strategy to mitigate Microbiologically Influenced Corrosion (MIC) was successfully established. This approach transitions from conventional fixed-interval dosing to targeted treatments tailored to pipeline segment risk classifications (Low, Moderate, High, Critical), with a focus on Main Oil Lines (MOLs) and associated transfer lines. Key elements of the strategy include:

- **Low-Risk Segments:** Reduction or suspension of biocide dosing to minimize unnecessary chemical usage and environmental impact.

- **High and Critical-Risk Segments:** Enhanced treatment involving extended biocide contact times, increased dosages (up to 50%), and formulation adjustments customized to microbial profiles. This was applied to the main oil line, considering the minimum pump capacity and dose rate in relation to the water phase. This consideration ensures the biocide is effectively distributed and in contact with the water phase for the required time.
- **Microbial Monitoring:** The strategy includes the proactive deployment of advanced diagnostics such as quantitative Polymerase Chain Reaction (qPCR), Adenosine Triphosphate (ATP) assays, and culture-based methods. Increase monitoring frequency up to weekly or biweekly monitoring with appropriate sampling points and preservation.

This strategy enables optimized chemical usage while delivering effective MIC control tailored to segment-specific risk profiles. It establishes a scalable, adaptive framework that significantly enhances biocide efficiency, advances predictive MIC management, and supports environmental sustainability across upstream pipeline network. The approach has achieved approximately USD 4.5 million in annual OPEX savings and an 81% reduction in biocide-related expenses, while maintaining robust microbial and corrosion control. Inspection data, microbial trend analysis, and real-time operational feedback drive the continuous refinement of the program.

Table 4—Optimized Risk-Based Biocide Treatment Summary

Item	Transfer Lines Treatment		MOL Treatment	
	Existing	Optimized	Existing	Optimized
Dose Rate (ppm) & Contact Time	500 ppm (3 hrs)	No Treatment	500 ppm (3 hrs)	200 ppm (5 hrs) Two MOLs at low risk: No Treatment
Treatment Frequency	Monthly	NA	Monthly	Monthly
Treatment Triggers (Serial dilution Test)	SRB counts > 100 cells/mL	Biocide reactivated if > 100 cells/mL detected	SRB counts > 100 cells/mL	Same

Conclusions and Strategic Recommendations

This study highlights the critical importance of implementing a risk-based, data-driven framework for the effective management of Microbiologically Influenced Corrosion (MIC) in upstream oil and gas pipeline systems. The development and application of a MIC Risk Matrix, calibrated with extensive historical integrity and operational data, has facilitated the systematic classification of pipeline segments by their susceptibility to microbial corrosion. This stratification has proven instrumental in prioritizing inspection efforts, refining monitoring accuracy, and optimizing mitigation measures, thereby enhancing resource allocation efficiency and improving overall asset integrity.

This study underscores the importance of a **risk-based, data-driven framework** for managing Microbiologically Influenced Corrosion (MIC) in upstream pipelines. One of the most significant achievements of this initiative is the successful optimization of biocide treatment protocols, tailored according to MIC risk profiles. By integrating advanced microbial diagnostic techniques such as quantitative polymerase chain reaction (qPCR) and Most Probable Number (MPN) assays, treatment regimens have been refined to achieve superior efficacy, cost-effectiveness, and environmental stewardship. This success should provide reassurance and confidence in the effectiveness of the strategies. Notable outcomes include:

- Reduced biocide dosing frequency and concentration in low-risk segments without compromising corrosion protection.

- Enhanced biocidal performance in high- and critical-risk zones, demonstrated by significant suppression of sulfate-reducing bacteria (SRB), nitrate-reducing bacteria (NRB), and total recoverable bacteria (TRB), alongside effective inhibition of biofilm formation.

Microbiological Monitoring and Sampling Trends

Effective MIC management combines routine planktonic and sessile sampling with corrosion assessment (TM0212, 2018). Key points include:

- **Planktonic Sampling:** Free-floating bacteria sampled at critical locations to support trend analysis and KPI evaluation.
- **Sessile Sampling:** Biofilm-forming bacteria monitored via corrosion coupons or bioprobes, flush-mounted at the line bottom for at least one month, with counts tracked alongside general and pitting corrosion rates.
- **KPIs and Data Interpretation:** Microbial data must be interpreted with corrosion measurements to define system-specific thresholds, considering analytical method differences (e.g., SRB counts via serial dilution vs. qPCR) and chemical indicators (sulfide, sulfur deposits, organic acids), while corrosion signals include coupon weight loss, pit depth, ER/LPR probe rates, dissolved iron (Fe^{2+} /total), and ILI/UT inspection trends.
- **Operational Factors:** MIC risk assessment requires accounting for water cut, flow velocity, fluid composition, temperature, pressure, and treatment regimes. Routine monitoring validates the effectiveness of mitigation strategies and reductions in pipeline damage.

This integrated approach demonstrates that combining MIC risk modeling, targeted monitoring, and optimized mitigation can effectively safeguard pipeline integrity while improving operational efficiency.

Strategic Recommendations

To sustain and advance the MIC management framework, the following strategic actions are advised. These recommendations are not just suggestions, but they are crucial for the continued success of the MIC management framework. Their implementation is urgent and necessary.

1. **Expand and Update the Risk-Based MIC Framework:** Continuously integrate real-time inspection data, microbial diagnostics, and operational metrics to ensure adaptive and asset-specific risk assessments. This means that risk assessments will be tailored to the specific conditions and characteristics of each asset, allowing for more accurate and effective mitigation strategies.
2. **Implement Condition-Based Biocide Protocols:** Adopt flexible dosing strategies that are responsive to fluctuating microbial populations and environmental conditions, thereby reducing overtreatment while maintaining adequate protection.
3. **Enhance Microbial Surveillance:** Broaden routine monitoring to systematically include SRB, TRB, and NRB populations, employing high-resolution diagnostics (e.g., qPCR, MPN) for early detection and risk anticipation.
4. **Strengthen Baseline Data Collection:** Improve consistency and detail in microbial and operational data to better characterize MIC drivers and emerging microbial threats.
5. **Develop Responsive Escalation Protocols:** Define microbial thresholds and corresponding intervention strategies to enable timely escalation of mitigation efforts when critical risk levels are exceeded.
6. **Engage in Industry Benchmarking and Collaboration:** Maintain active dialogue with peer operators and industry bodies to validate methodologies, adopt new technologies, and align with global best practices.

7. **Incorporate MIC Mitigation in Asset Design:** Integrate MIC considerations such as biocide injection infrastructure and diagnostic capabilities—into early design and commissioning phases to safeguard long-term integrity.
8. **Institutionalize Periodic Review and Optimization:** Regularly evaluate microbial trends, biocide effectiveness, and inspection outcomes to drive continuous improvement and adapt to evolving operational challenges. This emphasis on continuous improvement should make the audience feel that the strategies are always evolving and improving. To improve the reliability of serial dilution SRB quantification, it is recommended that:
 - Sampling procedures adhere strictly to microbiological best practices to preserve sample integrity.
 - Sampling maintains trend continuity to facilitate meaningful temporal comparisons.
 - Samples should be collected from the 6 o'clock position of pipelines, where microbial activity is typically highest due to fluid and sediment accumulation.

Adaptive treatment design refers to the ability to adjust biocide dosing and application methods based on real-time data and changing conditions, ensuring that the treatment remains effective and efficient. This approach, combined with operational agility, allows for timely and appropriate responses to emerging threats and operational changes.

Acknowledgments

The authors sincerely thank Abu Dhabi National Oil Company (ADNOC) Onshore and TotalEnergies for their invaluable support, technical guidance, and collaboration throughout the course of this work.

References

- Bonis. (2015). H₂S + CO₂ Corrosion: Additional Learnings From Field Experience. *Corrosion* 2015.
- Bruijnen. (2020). Integrated Approach toward diagnosing microbially induced corrosion in the Petroleum Industry. *SPE Production Operations, SPE* 197059.
- Chen, L. W. (2021). Effect of Sulfate-Reducing Bacteria (SRB) on the Corrosion of Buried Pipe Steel in Acidic Soil Solution. *Coatings*, **11**.
- Crolet, J.-L. (2005). Microbial Corrosion in the Oil Industry: A corrosionist's View. In J.-L. Crolet, *Petroleum Microbiology* (pp. 143–169). Washington DC: ASM Press.
- El-Sherik. (2017). Trends in oil and gas corrosion research and technologies: Production and Transmission. *Woodhead Publishing Engineering Core EBSCOhost*.
- Kermani Bijan, D. H. (2019). Corrosion and Materials in Hydrocarbon Production. In *Chapter 11, Microbiologically Influenced Corrosion (MIC)*. New York: ASME Press.
- King. (1972). Corrosion of mild steel by ferrous sulphide. *British Corrosion Journal*, 41–45.
- Lee, C. (1993). Corrosion of mild steel under anaerobic biofilm. *Corrosion Science*, 186–199.
- NACE 31205, I. (2006). Selection, Application, and Evaluation of Biocides in the Oil and Gas Industry. Houston, TX: 31205, Technical Committee Report (Task Group 075, STG 31).
- SP0106, A. (2018). Control of Internal Corrosion in Steel Pipeline and Pining systems. *AMPP*.
- Tiratsoo. (2013). Pipeline pigging and Integrity Technology 4th. *Clarion Technical Conference LLC*.
- TM0194, N. (2014). Field Monitoring of Bacterial Growth in oil and Gas Systems. *NACE International*.
- TM0212, N. I. (2018). Detection, testing, and evaluation of microbiologically influence corrosion on internal surfaces of pipelines. *NACE International*.